Fast Flood Extent Monitoring With SAR Change Detection Using Google Earth Engine

Ebrahim Hamidi, Brad G. Peter, David F. Muñoz, Hamed Moftakhari, and Hamid Moradkhani

***Abstract*— Flooding is one of the most frequent and disastrous natural hazards triggered by extreme precipitation, high river runoff, hurricane storm surges, and the compounding effects of various flood drivers. This study introduces a new multisource remote sensing approach that leverages both multispectral optical imagery and the weather- and illumination-independent characteristics of synthetic aperture radar (SAR) data to streamline, automate, and map geographically reliable flood inundation extents. Utilizing the near real-time and cloud computing capabilities of Google Earth Engine (GEE), this process facilitates data acquisition and enables large-scale flood monitoring in an expeditious manner. Two major hurricanes along the U.S. Gulf Coast were evaluated: 1) the 2021 Hurricane Ida to the south of New Orleans, LA, USA, and 2) the 2017 Hurricane Harvey to the east of Houston, TX, USA. We devised a change detection and thresholding framework using multitemporal SAR imagery and validated the results with flood extent maps derived from Landsat 8 and Sentinel-2 optical imagery. We demonstrate that constant threshold values for flood extraction from SAR change detection indices are not ubiquitously suitable for all geographies; thus, we outline a heuristic that can be used to select thresholds suitable for specific sites through a fully automated sensitivity analysis. The results indicated high agreement between the SAR and optical imagery (77%–80%), with SAR providing the benefit of under-cloud detection. Furthermore, our results contribute to scaling the SAR approach to produce rapid and accurate information for decision-makers and emergency responders during time-sensitive flood events.**

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I. Introduction

CLIMATE change has a great impact on our environment and is known to contribute to an increase in extreme natural events [1], [2]. Particularly, flooding has been regarded as one of the most frequent and destructive natural hazards in recent decades [3], [4] due to precipitation [5], [6], river runoff [7], [8], hurricane storm surges [9], [10], and compound effects of various flood drivers [11], [12], [13].

Coastal areas, due to the densely populated residence, economic activity, recreational opportunity, and military operations, are considered strategic and desirable places to live and conduct business [14], [15], yet these areas are prone to multiple natural disasters [11]. Hurricanes are among the most powerful and deadliest natural hazards, and the main drivers of coastal flooding in recent decades [16], [17]. The Gulf of Mexico is one of the most vulnerable coasts in the world and has been hit by hurricanes with increasing frequency since 2004 [18], [19]. For instance, the 2005 Hurricane Katrina that struck the Gulf Coast of the U.S. caused more than 1833 deaths [20]; the cost of damages due to Hurricane Katrina was approximately $160 billion [21]. The 2017 Hurricane Harvey was another recent hazardous and costly hurricane along the Southeastern coast of the United States. Harvey moved slowly with heavy rainfall, which exacerbated the large-scale devastating flood. It took a toll of more than a hundred human lives (68 diect deaths) and resulted in more than $125 billion of damage [22], [23], [24], [25], [26], [27]. The recent powerful Category 4 Hurricane Ida in the Gulf Coast made landfall on August 29, 2021, near Port Fourchon, LA, USA. It caused life-threatening floods and damaged essential urban infrastructure; nearly one million residential houses and businesses along the Gulf Coast were affected [28], [29]. According to the Centers for Disease Control and Prevention (CDC) report, Hurricane Ida is regarded as the fourth most deadly hurricane since 2012 in the United States (https://stacks.cdc.gov/view/cdc/110013).

Recent and recurring hazardous floods highlight the pressing need for state-of-the-art procedures for delineating floods in the affected areas [30]. Rapid and accurate methods to delineate flood extent are crucial for decision-makers and emergency planners to provide effective event-response

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measures that can help save human lives and identify urban areas prone to infrastructural damage. Earth-observing satellite data are considered among the most useful and applicable tools that enable geoscientists to access accurate and near real-time data resources with high spatial and temporal resolutions.

Optical satellites, such as the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), the Satellite for Observation of Earth (Satellite Pour l'Observation de la Terre, SPOT), the Moderate Resolution Imaging Spectroradiometer (MODIS), the Advanced Very High Resolution Radiometer (AVHRR), and the Land Remote-Sensing Satellite (Landsat), have been collecting image data for several decades. In addition, several methods have been devised to leverage multispectral images for extracting water and nonwater features, e.g., multispectral image classification, visual interpretation, decision rule classification, band rationing, logical filtering, and object-based classification [3], [31], [32]. Furthermore, various indices, such as the normalized difference vegetation index (NDVI), the normalized difference water index (NDWI), the modified NDWI (MNDWI), and the normalized differential pond index (NDPI), have also been widely used for detecting water bodies in optical imagery [4], [33], [34], [35]. Although optical satellites have been providing imagery since the 1970s, these image data are highly affected by meteorological and diurnal cycles as they depend on solar reflectance from the Earth's surface to capture land and water features.

In recent years, synthetic aperture radar (SAR) satellites have been used as an alternative for flood detection and monitoring since radar data provide researchers with high-resolution and near real-time surface data, regardless of weather conditions and time of day [3], [22], [36]. SAR data are very useful for flood detection as demonstrated in the literature through numerous classification methods. This is because flooded areas appear dark in SAR images due to a low backscatter signal on water bodies/flooded areas compared to other land-cover classes [1], [37]. Several methods have been proposed to map floods using SAR data, such as supervised classification, backscatter threshold algorithms, support vector machine (SVM), fuzzy logic classification, region growing, and texture analysis methods [3], [4], [38], [39]. The majority of these techniques utilize single-date imagery for flood detection even though multitemporal methods that consider both preflood as a baseline and postflood SAR data can provide more reliable flood maps [40], [41], [42], [43], [44], [45].

SAR change detection is a commonly used image analysis method that uses imagery from different dates to identify changes across a landscape [46]. This method has been used for a wide range of applications, including natural resource surveys [47], urban planning [48], [49], freight traffic surveillance [50], flood mapping [3], [22], [36], [41], [51], [52], agricultural monitoring [4], [53], and earthquake damage assessment [54]. SAR change detection methods generally have two primary components: 1) generating difference images and 2) classifying the difference images using supervised or unsupervised techniques [55], [56], [57]. The unavailability of reliable ground-truth information and complexity of classifying SAR images to various land-cover categories often makes unsupervised methods more suitable for SAR image change detection [46], [58], [59], [60]. Accordingly, several statistical techniques have been proposed to determine unsupervised threshold values for SAR imagery categorization, such as the generalized Kittler and Illingworth thresholding algorithm [61], the expectation–maximization algorithm [60], and the Bayesian network algorithm [62], [63]. These methods have been widely utilized for flood extent extraction [58], [59], [60].

Nevertheless, flood detection and monitoring with SAR data do present some challenges [64]; for instance, radar shadows or layover due to buildings and taller vegetation could incorrectly classify dry areas as wet ones [65], or wind and heavy rainfall can temporarily shape a roughness on the water surface and hinder flood delineation [66]. In addition, the double-bounce backscatter signal of flood under vegetation or in high-intensity urban areas appears very bright (high backscattering radar signal) in SAR data [40], [67], [68]. However, the advantages of SAR data outweigh its disadvantages and remain an invaluable data type for flood detection [4].

Single-source remote sensing data are often not ideal for emergency flood mapping. Limitations such as satellites' temporal resolution (revisiting period), spatial resolution, dependence on atmospheric weather conditions and solar reflectance variation, and even sensor errors and defects make remote sensing imagery susceptible to errors. However, urgent disaster management is key during extreme floods, and it cannot wait for perfect high-quality images. In this circumstance, even suboptimal or defective satellite images are beneficial resources for flood monitoring; however, multisource methods can overcome some of the inefficiencies of single-source methods [69], [70]. Using multisource optical imagery/SAR provides multiple lines of evidence that enables highly accurate water extraction. Optical imagery has been vetted over many years, and SAR imagery adds the benefit of all-weather and day-night operation capabilities [69]. Combining multitemporal and multisource remote data for flood inundation analysis is a new advancement that has been used for different applications [65], [71], [72], [73], [74]; however, there is more research and tools needed to provide fast, fully automatic, and high-precision floodwater maps [69].

Recently, by leveraging cloud computing architectures and the advantages of high-performance computing (HPC), geoscientists have access to a wide range of Earth Observation data. This revolution has facilitated the processing large volumes of data in HPC cloud platforms and the rapid production of map products. These tools also reduce the need to download data and outfit a workspace that can move and process global data with costly specialized hardware and software [75], [76]. Google Earth Engine (GEE) has had a pioneering role in providing an HPC cloud computing platform and housing a comprehensive and continually updated archive of remote sensing data that make it possible to develop local- and global-scale satellite

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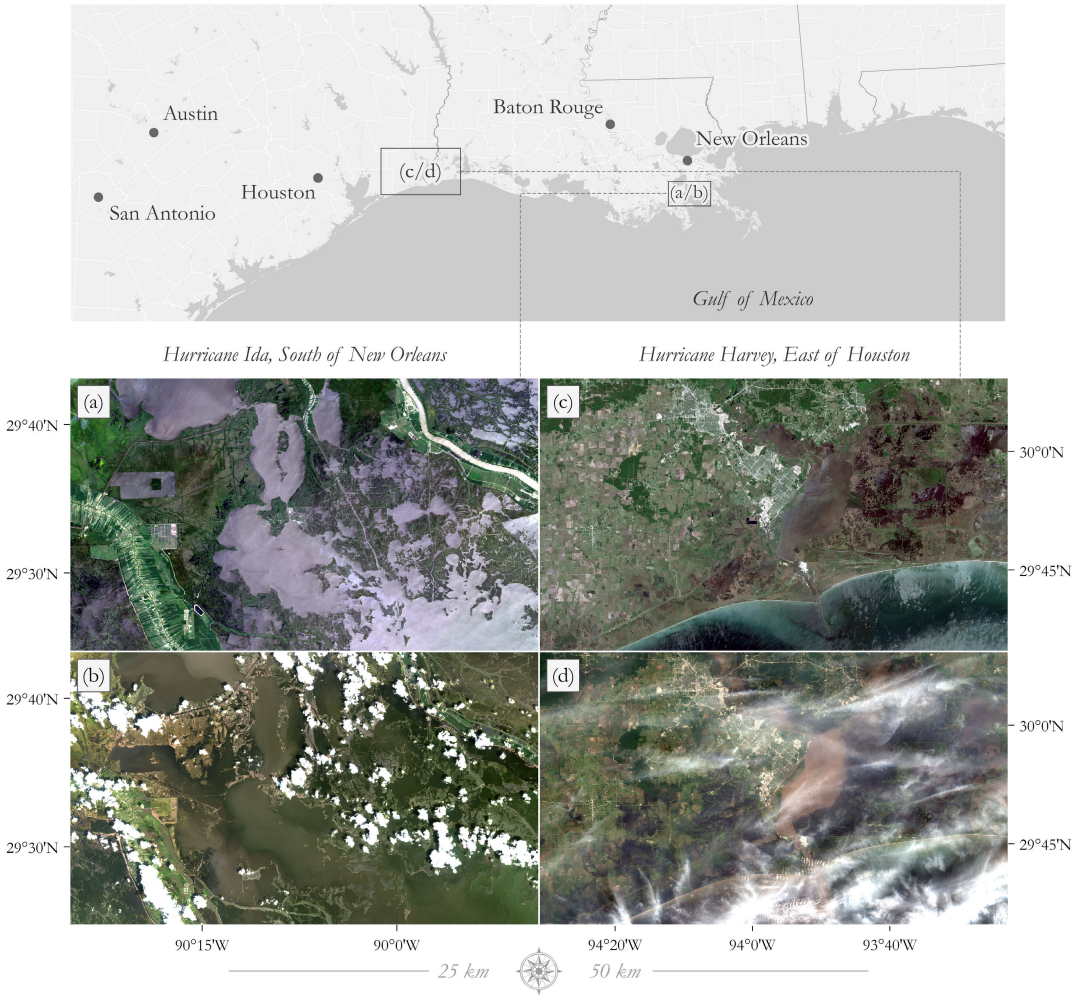


Fig. 1. Location of the study areas. The region south of New Orleans (a) before and (b) after the Hurricane Ida flood. The region east of Houston (c) before and (d) after the Hurricane Harvey flood.

data analytics and interpretation [4], [22], [75]. Notable research leveraging the capabilities of GEE includes forest land-cover investigation [77], surface water monitoring [78], cropland and soils land-cover observation [79], species suitability [80], [81], flood depth estimation [82], and drought tracking [83], [84].

The objective of this study is to analyze and develop tools for near real-time flood monitoring through a novel combining of multitemporal and multisource remote sensing data. We use an SAR change detection and thresholding method, and apply sensitivity analytics and thresholding calibration, using SAR- and optical-based indices in a format that is streamlined, reproducible, and geographically agile. We leverage the massive repository of satellite imagery and planetary-scale geospatial analysis tools of GEE to devise a flood inundation extent model that is both scalable and replicable. The flood extents from the 2021 Hurricane Ida and the 2017 Hurricane Harvey were selected to test the approach. The methodology provides a fast, automatable, and geographically reliable tool for assisting decision-makers and emergency planners during extreme flood events using near real-time multitemporal satellite SAR datasets.

II. Methods

A. Study Area

Recent hurricanes made the residents of the Gulf Coast of the United States vulnerable to extreme flood events [9]. Because of the frequent exposure of this area to coastal flooding due to hurricanes, we focus on two recent destructive hurricanes on the U.S. Gulf Coast and apply the proposed flood extent monitoring approach to the 2021 Hurricane Ida to the south of New Orleans and the 2017 Hurricane Harvey to the east of Houston (see Fig. 1). The city of New Orleans in the southeast of Louisiana was affected significantly by Hurricane Ida with intense winds, high storm surges, heavy rainfall, rivers runoff, and floods. In this research, we focus on the flood to the south of New Orleans, which covers approximately 2500 km^2 of land. Furthermore, intense rainfall and strong storm surges due to the slowly moving Hurricane Harvey resulted in devastating flooding in southeast Texas. We survey the flood to the east of Houston, around Sabine Lake and Port Arthur, where the 2017 Hurricane Harvey had its third landfall on August 30, 2017. This site covers approximately 7500 km^2 of land.

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TABLE I
LIST OF IMAGES USED FOR FLOOD MONITORING

Sensor	Date	Status	Orbit/Polarization	Spatial Resolution	Event
Sentinel-1	05-Aug-2021	Before Flood	Ascending / VH	10 m	Hurricane Ida
	17-Aug-2021				
	10-Sep-2021	After Flood			
	22-Sep-2021				
Landsat 8	30-May-2021	Before Flood	---	30 m	
	03-Sep-2021	After Flood			
Sentinel-1	01-Jun-2017	Before Flood	Ascending / VH	10 m	Hurricane Harvey
	13-Jun-2017				
	05-Sep-2017	After Flood			
Sentinel-2	24-Apr-2017	Before Flood	---	20 m	
	06-Sep-2017	After Flood			

The Landsat 8 imagery from May 30, 2021 [see Fig. 1(a)] shows the site to the south of New Orleans before the flood of the 2021 Hurricane Ida, and the image from September 03, 2021 [see Fig. 1(b)] depicts this area after the flood. By comparing these two images, flooded regions are clearly distinguishable, particularly in the center-left part of the images. As shown in Fig. 1(b), cloud cover in Landsat 8 image is a drawback of the optical imagery, which makes it difficult and sometimes impossible to use for flood mapping.

Fig. 1(c) and (d) shows the Sentinel-2 images before the flood (on April 24, 2017) and after the flood (on September 6, 2017), respectively, from the 2017 Hurricane Harvey, east of Houston. Problematically, most of the optical images after Hurricane Harvey for this region are covered by clouds; the least cloudy image is shown in Fig. 1(d) (about 50% of this image is covered by cirrus clouds; however, fortunately, most of the clouds cover the open water/ocean).

The average land elevation across the Texas site is 2.88 m with a standard deviation of 3.63 m, referring to the North American Vertical Datum of 1988 (NAVD88). Emergent herbaceous wetlands are the most prevalent land-use land-cover (LULC) across the area (31.51%), followed by open water (21.84%) and pasture/hay (21.52%). Cultivated crops and urban land (classes 21, 22, 23, and 24) occupy 4.64% and 9.67% of the area, respectively. In contrast to the Texas site, the average elevation of the Louisiana site is below sea level (−0.07 m) with a standard deviation of 0.48 m. Emergent herbaceous wetlands are the most prevalent LULC across the area (46.91%), followed by open water (40.95%) and pasture/hay (4.12%). Cultivated crops and urban land occupy 0.71% and 2.46% of the area, respectively. Elevation data were collected from The U.S. Geological Survey (USGS) 3DEP and LULC from USGS NLCD 2019 (https://www.usgs.gov/data/national-land-cover-database-nlcd-2019-products).

B. Satellite Data

Sentinel-1 is a European Space Agency (ESA) radar imaging mission that began in October 2014 and provides continuous dual-polarization C-band SAR data with a temporal resolution of six or 12 days (i.e., each satellite has a 12-day revisit rate, resulting in a temporal resolution of six days when coupling both satellites in the constellation), three spatial resolutions (10, 25, and 40 m), and four band combinations (single vertical–vertical (VV) band or horizontal–horizontal (HH) band, and dual-band VV + VH and HH + HV). Also, there are four modes of SAR imagery available in Sentinel-1 data collection: stripmap (SM), interferometric wide (IW) swath, extra wide (EW) swath, and wave (WV) [85].

Sentinel-1 Level-1 Ground Range Detected (GRD) imagery is available in GEE ("COPERNICUS/S1_GRD"), and this image collection has already been preprocessed by: 1) updating orbit metadata and removing GRD border (e.g., low-intensity noise and invalid data) and thermal noises (e.g., additive noise in subswaths); 2) conducting terrain correction and radiometric calibration of backscatter intensity using sensor calibration parameters; and 3) transforming backscatter coefficients into decibel scale (https://developers.google.com/earth-engine/guides/sentinel1). The latter has been also reported in other studies [37], [86]. Furthermore, we applied a smoothing filter to the Sentinel-1 radar imagery to reduce the speckle-effect [87], [88].

We leverage the benefits of Sentinel-1 radar imagery data (i.e., all-weather and day-and-night reliability) available in GEE for flood extent delineation in coastal areas. Although Sentinel-1 is a new Earth observer satellite (start date October 3, 2014), its temporal resolution makes it possible to collect multiple images before and after flood events for change detection flood mapping methods [59], [81]. We use VH polarization, ascending pass direction, and IW-mode SAR imagery with a 10-m spatial resolution for conducting flood mapping in coastal regions (see Table I). Also, we filter steep areas with greater than 5° slope using the USGS 3D Elevation Program (3DEP) 10-m spatial resolution Digital Elevation Model ("USGS/3DEP/10m"; USGS 3DEP DEM). To remove permanent water features and isolate flooded areas, we use the European Commission's Joint Research Centre (JRC) Global Surface Water Mapping Layers, v1.1 ("JRC/GSW1_0/GlobalSurfaceWater"). This imagery provides

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30-m spatial resolution imagery of surface water distributions derived from 32 years (1984–2015) of Landsat data [82].

In addition to SAR data, we use the USGS Landsat 8 Surface Reflectance Tier 1 dataset ("Landsat/LC08/C01/T1_SR") that provides orthorectified surface reflectance images since 2013. This atmospherically corrected dataset contains visible, near-infrared, short-wave infrared, and thermal infrared bands and has cloud, shadow, water, and snow quality control bands for masking pixel data. Sentinel-2 is another ESA satellite that supplies multispectral imagery with 13 spectral bands and a wide swath width of 290 km. It provides global coverage with a high spatial resolution (i.e., 10–60 m depending on the bands), which can be used for flood monitoring and emergency response. We utilize the Sentinel-2 image collection that is available during our study timeframe in GEE ("COPERNICUS/S2"). In this study, we utilize Landsat 8 and Sentinel-2 optical imagery to analyze the sensitivity of the SAR change detection approach to varying threshold values.

C. Multisource and Multitemporal Flood Mapping

In this section, we illustrate a method that combines multisource and multitemporal remote sensing datasets for highly accurate and efficient floodwater extraction. First, multitemporal SAR images are used to apply a change detection technique to calculate SAR index images and highlight inundated areas after a flood event including the normalized difference flood index (NDFI), the ratio image (RI), and the difference image index (DII). Then, optical images are used (even if they are affected by clouds, shadows, or stripes) to assess the accuracy and conduct an automated sensitivity analysis to find the optimum SAR index threshold values for flood extraction. Finally, using the optimum calculated thresholds, the final flood maps are extracted from SAR index images. Fig. 2 presents a conceptual diagram that outlines the steps taken in this method.

The change detection method is a reliable and straightforward technique for delineating flooded areas. This method has been used in several studies for fast processing and response to time-sensitive events [4], [40], [43], [46], [50], [53], [57], [59], [89], [90]. We retrieve a series of SAR images that capture preflood and postflood conditions, and use them as reference and flood image stacks, respectively. For this, we first establish a baseline by calculating the mean of backscattering signals from the reference image stack to identify where/when no flooding occurred. Then, we create postflood imagery by computing the minimum of backscattering signals from the flood image stack. Considering the minimum of backscattering signals of SAR imagery after a flood event can capture the maximum flood extent in the study region [89]. Finally, using the reference and flood image stacks, we calculate the following SAR-derived indices.

1) NDFI: NDFI highlights flooded areas [4], [89] and is calculated using the following equation:

$$\mathrm{NDFI} = \frac{|\mathrm{mean}(\sigma_{0_{[\mathrm{BF}]}})| - |\mathrm{min}(\sigma_{0_{[\mathrm{AF}]}})|}{|\mathrm{mean}(\sigma_{0_{[\mathrm{BF}]}})| + |\mathrm{min}(\sigma_{0_{[\mathrm{AF}]}})|} \tag{1}$$

where $\sigma_{0[\mathrm{BF}]}$ and $\sigma_{0[\mathrm{AF}]}$ are the backscatter coefficient of SAR imagery before and after flood, respectively.

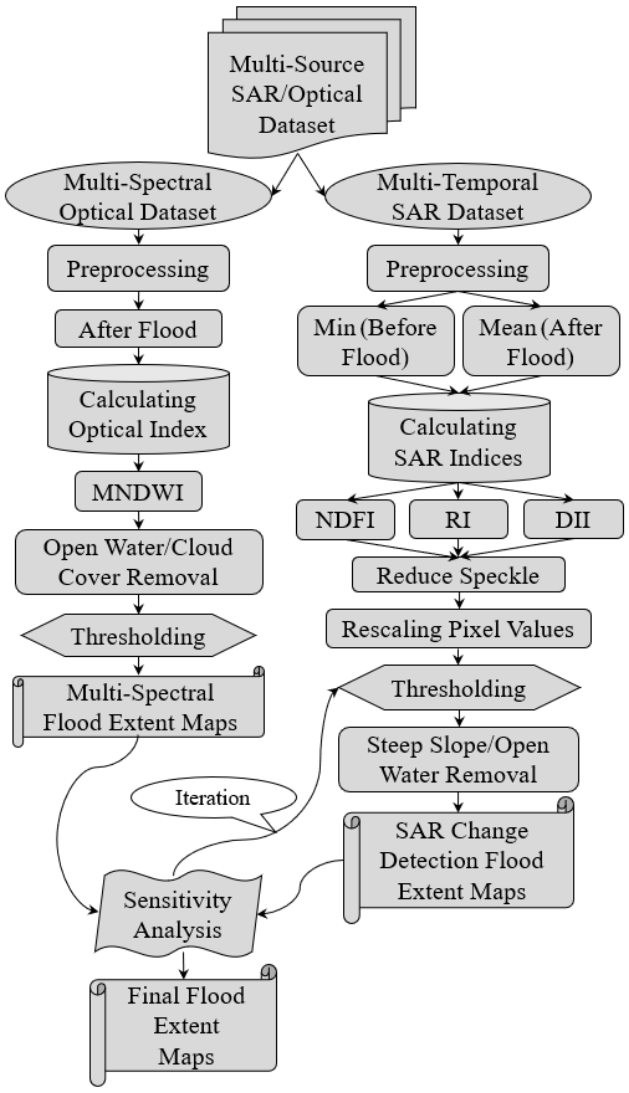


Fig. 2. Diagram of the multisource and multitemporal method for flood extent delineation using SAR and optical datasets in GEE.

2) RI: RI is widely used for flood delineation [3], [91]. To calculate RI, the minimum of absolute backscattering values after flooding is divided by the mean of absolute backscattering values of the imagery before flooding, as shown in the following equation:

$$\mathrm{RI} = \frac{|\mathrm{min}(\sigma_{0_{[\mathrm{AF}]}})|}{|\mathrm{mean}(\sigma_{0_{[\mathrm{BF}]}})|}. \tag{2}$$

3) DII: DII is the absolute difference between preflood and postflood images proposed by Long et al. [43]. However, in this study, the postflood image is the minimum backscattering value of the multitemporal SAR image stack after flooding, and the preflood image is the mean of backscattering values of the multitemporal SAR image stack before flooding, as shown in the following equation:

$$\mathrm{DII} = |\mathrm{min}(\sigma_{0_{[\mathrm{AF}]}})| - |\mathrm{mean}(\sigma_{0_{[\mathrm{BF}]}})|. \tag{3}$$

Threshold Value: The computed SAR index pixel values associated with no change (i.e., nonflooded) exhibit values

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substantially different from the changed (i.e., flooded) pixels; hence, analyzing the distribution of the values derived from the SAR index, which compares preflood and postflood, enables the identification and extraction of flooded pixels. One of the most widely used approaches for selecting the threshold values is based on this assumption that the density function (i.e., the distribution or histogram) will elucidate a mathematically significant difference between categories, in this case, flooded and nonflooded pixels [60]. This method is similar to the commonly used Otsu's method [92], as well as the Jenks natural breaks method in cartography, where pixels are parsed into categories based on differences inherent to the data distribution [93], [94]. The formula used to assign this threshold is based on the number of standard deviations (k_f) from the mean of the SAR index image, and the threshold value (Thr) can be determined by the following equation [4], [40], [43], [89], [90]:

$$\text{Thr} = \text{mean(SAR Index Image)} \pm k_f \times \text{stdev(SAR Index Image)}. \quad (4)$$

The minus sign was used for NDFI and the pixel values below the threshold value were considered flooded pixels. On the other hand, the plus sign in (4) was used for RI and DII so that pixel values greater than the threshold value were mapped as flooded pixels. In (4), the k_f coefficient is typically 1.5 by default, as it has been demonstrated as suitable in many similar studies [4], [40], [43], [89], [90]; however, we found that this constant thresholding value is not always suitable and would fail to extract the flooded pixels for different indices and geographies. Thus, we conducted a sensitivity analysis of the k_f coefficient to estimate the threshold value that resulted in the highest accuracy when extracting flooded pixels from the SAR index images. The histograms of each SAR index image were examined, and k_f values were tested iteratively ranging from 0.0 to 2.0 with an increment of 0.1 (21 tests total). The extracted flooded area from each k_f value was validated against the flood extent extracted from multispectral imagery using the MNDWI. The results of the sensitivity analysis on the SAR and optical flood extent maps provide us with the optimum k_f values that were ultimately used to produce the final flood extent map for each index and each geography. Flood extent maps obtained from the above multisource and multitemporal approach require some postprocessing modifications. First, permanent water land cover is masked out using the JRC Global Surface Water Mapping Layers. Then, the USGS 3DEP DEM was used to remove areas that have slopes greater than 5° where floods would be unlikely [89]. Finally, we reduced noise from the resulting imagery by applying a smoothing filter to reduce the speckle effect of radar imagery.

MNDWI: As part of the multisource data approach, we use available multispectral images of Landsat 8 and Sentinel-2 satellites for flood mapping. For extracting flooded pixels, we calculate MNDWI, which is commonly used in the literature for detecting water bodies in optical imagery [33], [35], [40], [95], [96], [97]

$$\text{MNDWI} = \frac{\text{Green Band} - \text{SWIR Band}}{\text{Green Band} + \text{SWIR Band}} \quad (5)$$

where Band 3 is the Green Band for both Landsat 8 and Sentinel-2 satellites (0.533–0.590 and 0.560/0.559 μm, respectively); however, short-wave infrared (SWIR) band is Band 6 in Landsat and Band 11 in Sentinel-2 (1.566–1.651 and 1.6137/1.6104 μm, respectively). As water bodies absorb SWIR radiation significantly, MNDWI is one of the state-of-the-art optical indices that are used for water identification in optical satellite image analysis [40]. For flood detection purposes, we removed permanent water bodies from the resulting MNDWI image to assess the flooded area only. The code and methodology associated with this manuscript can be accessed via Harvard Dataverse [98].

D. Result Assessment and Sensitivity Analysis Procedure

In order to assess the performance of SAR flood maps, (6) estimates the agreement (Agr) percentage of SAR indices with respect to optical imagery results

$$\text{Agr\%} = B/O \times 100. \quad (6)$$

where B is the intersection area of the optical and SAR flood detections (both SAR and optical) and O is the area of the optical flood detections. This equation effectively shows the percentage of SAR flood detection that intersects with optical water detection. On the other hand, (7) calculates the total agreement (ToAgr) percentage according to SAR and optical estimated flood extent. We propose this equation to conduct a sensitivity analysis of flood extent maps resulting from the SAR- and Optical-based indices

$$\text{ToAgr\%} = B/(O + \text{Src} - B) \times 100 \quad (7)$$

where Src is the area of the SAR flood detections excluding the flood under cloud cover in the optical image. We exclude cloud cover from the SAR flood extent map to make it comparable with the optical image flood extent. B is subtracted from the denominator so that the intersection area is not counted twice. This effectively shows the overall difference between the SAR and optical flood detection areas where there are no clouds. Equation (7) considers disagreements where SAR may inaccurately classify areas as the flood. In (6) and (7), a value of 100 indicates perfect agreement between SAR and optical flood area, and a value of 0 indicates the opposite. To highlight the ability of SAR to capture water bodies beneath the cloud layer that the optical imagery can miss, (8) estimates the under-cloud flood (UndCldFld) percentage

$$\text{UndCldFld\%} = (S - \text{Src})/S \times 100 \quad (8)$$

where S is the area of the SAR flood detections (no cloud removal).

III. Results

A. SAR Flood Mapping

To map coastal floods using change detection and thresholding methods, we use multitemporal SAR images

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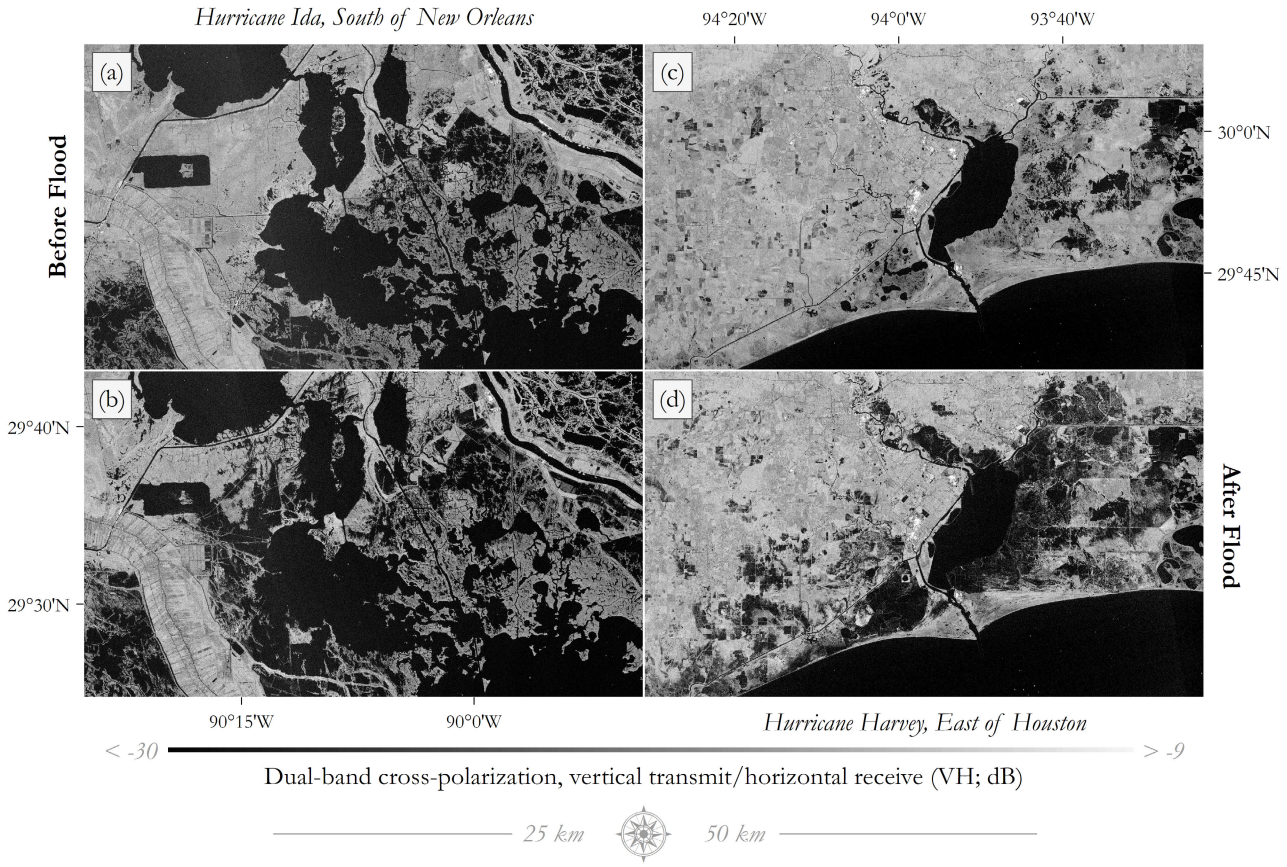


Fig. 3. Multitemporal SAR images before and after the flood. (a) Mean of multitemporal imagery (August 5 and August 17, 2021) before the flood. (b) Minimum of multitemporal imagery (September 10 and September 22, 2021) after the flood for the 2021 Hurricane Ida (south of New Orleans). (c) Mean of multitemporal SAR imagery (January 1 and January 13, 2017) before the flood. (d) SAR image of September 5, 2017, after the flood for the 2017 Hurricane Harvey (east of Houston).

before and after the flood. Fig. 3(a) depicts the mean of the backscattering signal of the reference image stack before the flood (August 5 and August 17, 2021), and Fig. 3(b) shows the minimum of the backscattering signal of the flood image stack after the flood (September 10 and September 22, 2021) of the 2021 Hurricane Ida (south of New Orleans) (see Table I). Comparing Fig. 3(a) and (b) clearly highlights new black regions (flooded areas) due to low backscatter signals. Similarly, Fig. 3(c) shows the reference image before the 2017 Hurricane Harvey, which is the mean of backscatter signals of SAR images stacked before the flood to the east of Houston (June 01 and January 13, 2017). For this study area, there were not sufficient multitemporal images after the flood in the GEE; therefore, we used the single-image backscatter signal of September 5, 2017, as an after-flood image [see Fig. 3(d)]. This SAR imagery is captured one day before the Sentinel-2 imagery (Fig. 1(d); on September 6, 2017), which has the closest date to the third landfall of Hurricane Harvey (on August 30, 2017) near Sabine Lake/Port Arthur (see Table I).

NDFI, RI, and DII are calculated using (1)–(3); then, pixel values of these SAR indices are rescaled between 0 and 1. The resulting NDFI, RI, and DII images are illustrated in Fig. 4(a)–(c) for the 2021 Hurricane Ida (south of New Orleans) and Fig. 4(d)–(f) for the 2017 Hurricane Harvey (east of Houston). As shown in Fig. 4, the changes from no-flood to flood land cover are highlighted in black for the NDFI image and in white for RI and DII images.

To map the flooded pixels from the SAR indices images, we use (4) and apply the thresholding criteria on the SAR indices images to extract flooded pixels. To calculate threshold values, we consider the k_f coefficient equal to 1.5 in (4) and extract the flooded pixels; this coefficient is a convention that has been applied in several previous studies [4], [40], [43], [89], [90]. This method returned threshold values of 0.5063, 0.0135, and 0.6003, which were used to delineate flood extents in the NDFI, RI, and DII images [see Fig. 5(a)–(c)], respectively, for the 2021 Hurricane Ida to the south of New Orleans. As illustrated in Fig. 5(a)–(c), the extracted flood maps using NDFI, RI, and DII indices correspond with the highlighted land cover changes in Fig. 4(a)–(c).

Likewise, for extracting flooded pixels for the 2017 Hurricane Harvey to the east of Houston, we consider $k_f = 1.5$ in (4) and survey the flood extent results. However, we found that this value was not sufficient to extract flood pixels in this region. Fig. 5(d)–(f) shows the extracted flood extent using NDFI, RI, and DII after applying threshold values of 0.3719, 0.0034, and 0.6944, respectively, for Hurricane Harvey. These figures depict that these threshold values do

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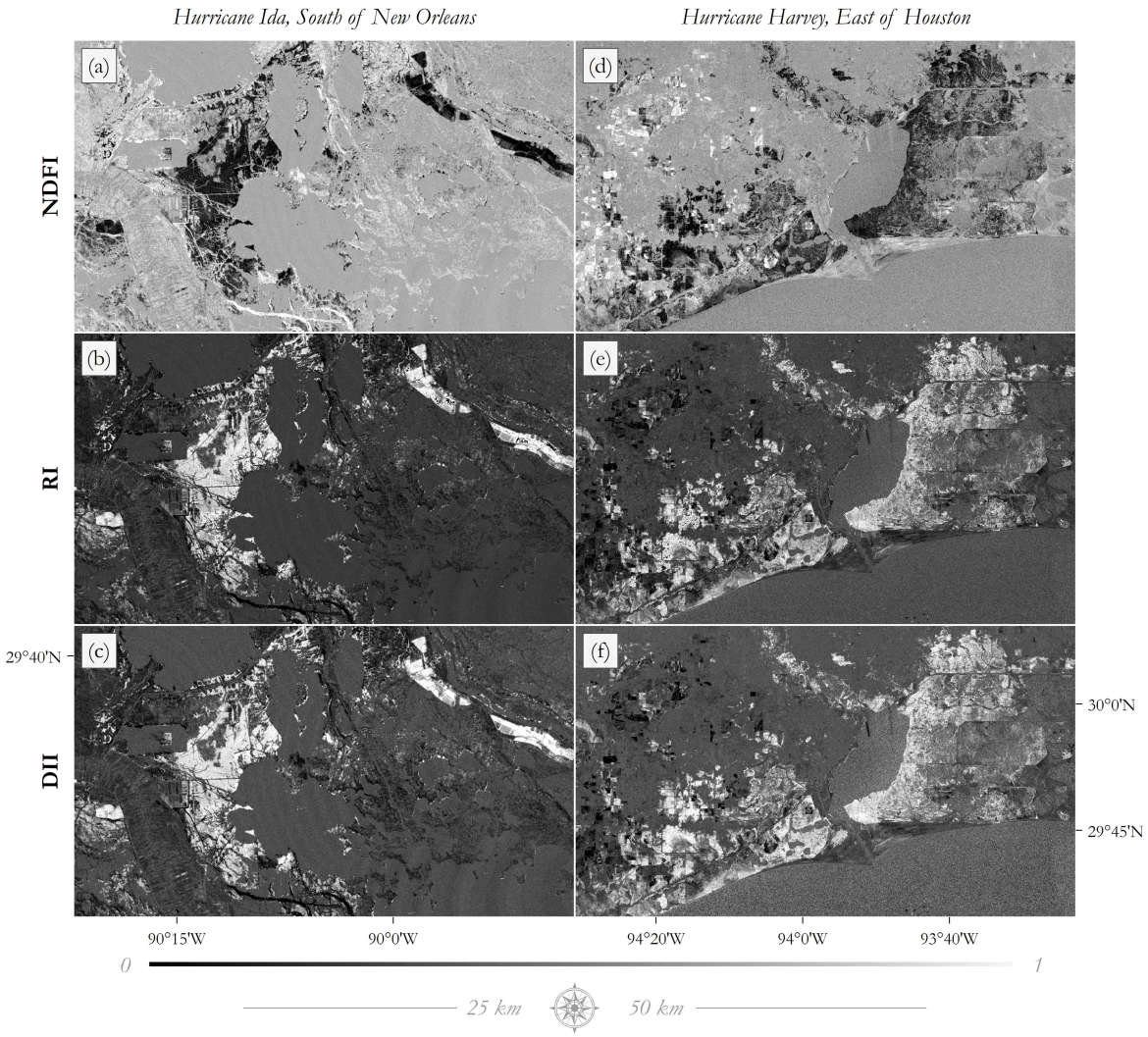


Fig. 4. Rescaled (a) NDFI, (b) RI, and (c) DII images for the 2021 Hurricane Ida (south of New Orleans). Rescaled (d) NDFI, (e) RI, and (f) DII images for the 2017 Hurricane Harvey (east of Houston). Land cover changes due to floods are highlighted in black for NDFI and white for RI and DII.

not extract the flooded area in the RI image [see Fig. 5(e)]; also, the extracted flood maps using the NDFI and DII images have a low visual correspondence between the highlighted areas in Fig. 4 and the extracted flood areas in Fig. 5 for Hurricane Harvey (right column of the figure). It is apparent that a predefined threshold may not guarantee a reliable flood extent extraction for different SAR indices, and calibration criteria may be required to estimate proper threshold values. Moreover, thresholds devised from any particular study may not be generalizable across geographies; thus, we elected to construct alternative thresholds that were more effective for the site to the east of Houston (Hurricane Harvey flood event).

B. Optical Imagery Flood Extent Estimation

For the calibration of threshold values, we use optical images and compare the SAR-based flood extent results with Landsat 8 and Sentinel-2 optical imagery-based flood extent. We calculate MNDWI using (5) for identifying water bodies and extracting floods from multispectral images. Then, we removed permanent water bodies using the JRC Global Surface Water Mapping Layers to highlight floods separately from the complete water footprint [99]; clouds and cloud shadows are also masked out. Threshold values of 0.5 and 0.4 correspond with flooded pixels from the calculated MNDWI layers for Hurricane Ida (Landsat 8) and Hurricane Harvey (Sentinel-2), respectively. This usage, i.e., selecting a threshold from MNDWI to identify water, is consistent with other studies (e.g., Baig et al. [33], Phiri et al. [100], and Ferral et al. [101]). Baig et al. [33] found that a threshold of 0.41 worked for identifying water in a lower basin of the Indus River, and Ferral et al. [101] identified thresholds for different forms of water/mixed-water (MNDWI $>$ 0.5 for L8-OLI) in an Argentine wetland. The result images of applying these steps on the after-flood images [see Fig. 1(b) and (d)] are presented in Fig. 6(a) and (c) for Hurricane Ida and Hurricane Harvey, respectively. Also, the extracted flooded areas using MNDWI images are highlighted in red in Fig. 6(b) and (d). These optical imagery-based MNDWI flood extent layers were utilized to conduct a sensitivity analysis procedure and calibrate the threshold values for the SAR

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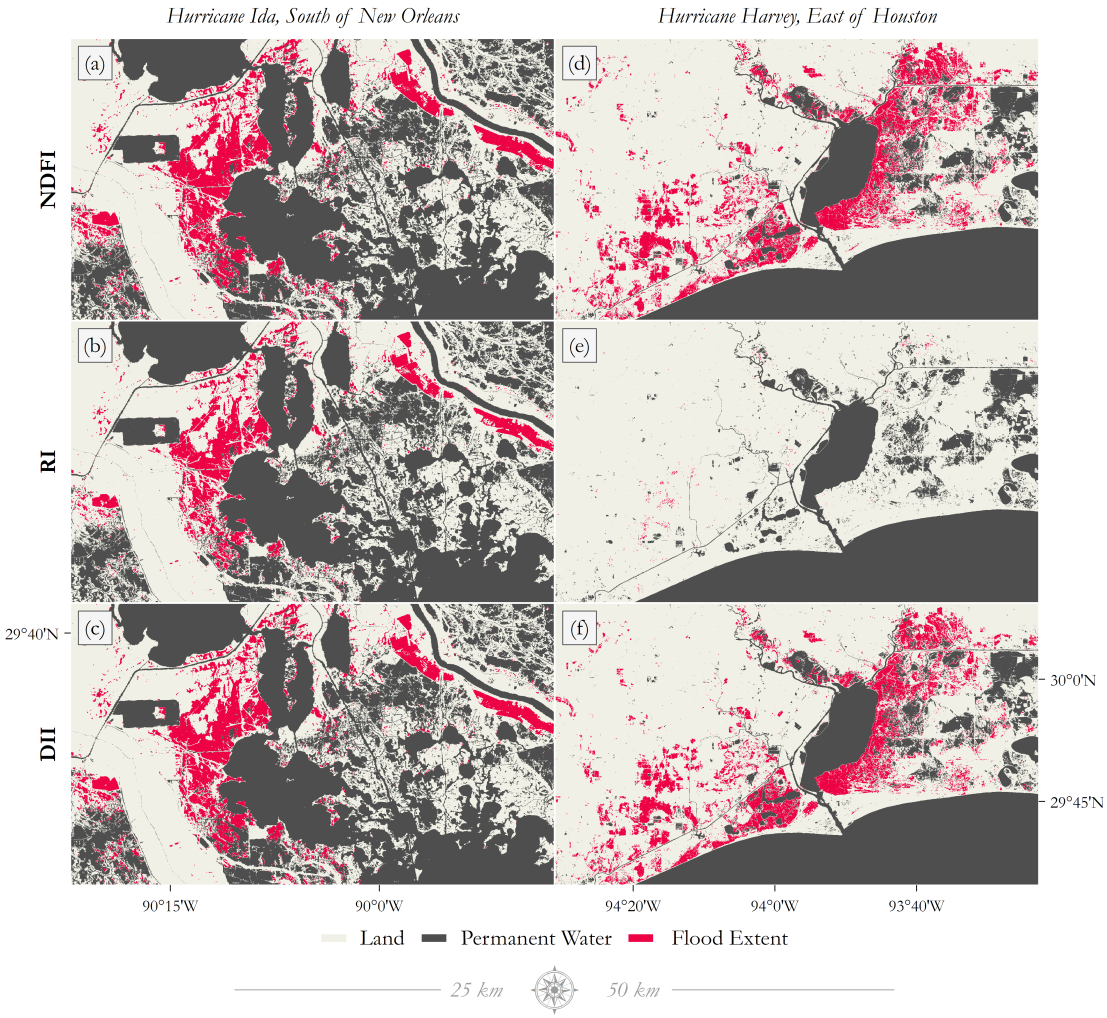


Fig. 5. Estimated flood extent by applying (a) threshold = 0. 5063 on NDFI, (b) threshold = 0.0135 on RI, and (c) threshold = 0.6003 on DII for the 2021 Hurricane Ida (south of New Orleans). Also, estimated flood extent using (d) threshold value equal to 0.3719 on NDFI, (e) threshold value equal to 0.0034 on RI, and (f) threshold value equal to 0.6944 on DII for the 2017 Hurricane Harvey (east of Houston).

change detection method. Although the optical imagery failed to delineate the flood where the imagery is covered by clouds, it is still helpful for estimating the extent of flooding across the area to relate the MNDWI to the SAR imagery.

C. Sensitivity Analysis and Threshold Calibration Procedure

Finding a threshold value for the change detection method is not a straightforward process and can be a challenge [40]. In this study, we initially considered $k_f = 1.5$ to compute threshold values and derived flood maps as suggested in similar studies [4], [40], [43], [89], [90]. Although the results were rather acceptable for the Hurricane Ida flood to the south of New Orleans, $k_f = 1.5$ failed to extract flooded pixels of Hurricane Harvey to the east of Houston (see Fig. 5). Fine-tuning of threshold values for SAR indices using optical flood extent and visual interpretation indicates that lower values for k_f can more adequately map flood extent in this study area. Then, using studies by Munasinghe et al. [102] and Wing et al. [103], who proposed useful criteria for validation procedures, we examine the results of (6) and (7) to evaluate the responsiveness of the SAR flood extent with varying k_f. Finally, we calibrate the thresholds for floodwater extraction from the SAR indices by relating them to the optical flood extent.

The histograms of pixel values of each rescaled SAR index (see Fig. 4) were examined, and the mean and standard deviation values were calculated. Then, threshold values equal to $\pm k_f$ standard deviations from the mean [formulation presented in (4)] were selected to delineate flooded pixels. The k_f coefficient was iteratively changed between 0.0 and 2.0 with an increment of 0.1. The calculated threshold values in each iteration are used to examine the sensitivity of the extracted flooded pixels from SAR indices images with an optical-based flood extent map using (7). The results of this procedure are presented in Fig. 7. This figure presents that the total agreement percentage is sensitive to k_f coefficient values, where low and high k_f values can lead to unsatisfactory agreements.

The optimum threshold value for each SAR index is selected using Fig. 7. Therefore, for mapping the 2021 Hurricane Ida flood, we consider $k_f = 1.6$ (Thr = 0.5013) for NDFI, which has the best total agreement percentage [see Fig. 7(a) and 8(a)]. Also, sensitivity analysis shows that $k_f = 1.00$ (Thr = 0.0119) is the optimal value to extract flood pixels from the RI image

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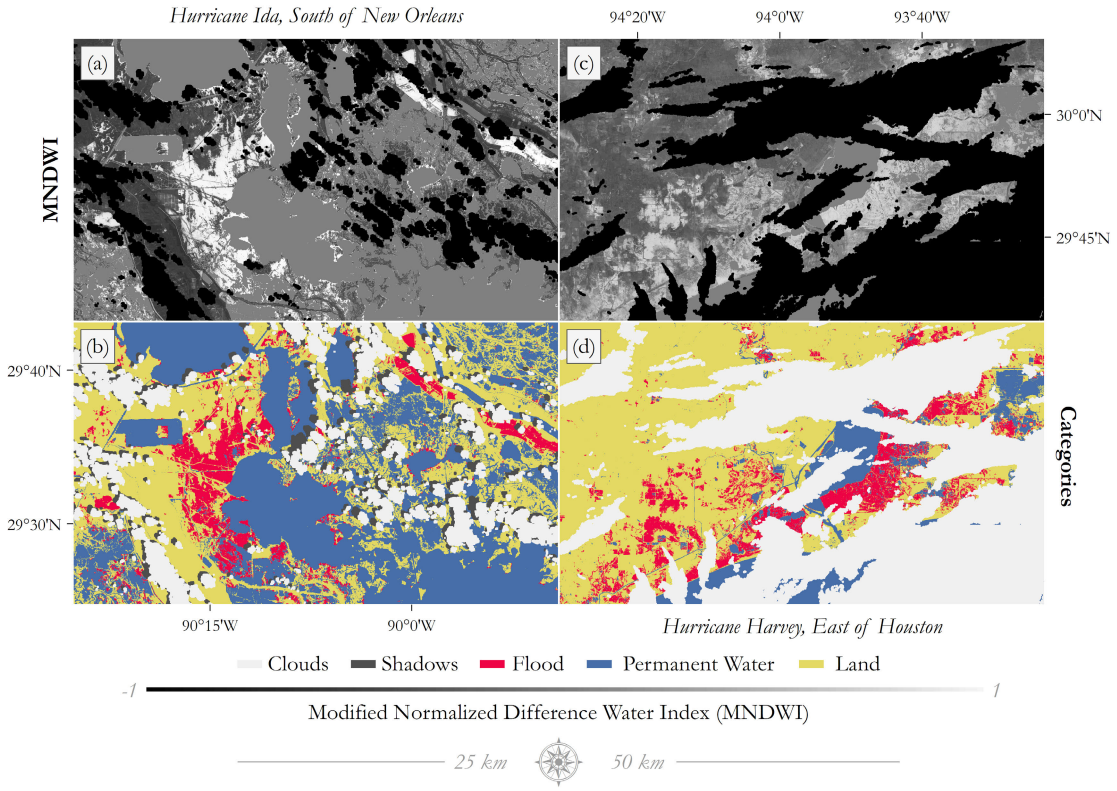


Fig. 6. (a) Computed MNDWI and (b) categorized MNDWI derived from Landsat 8 imagery after the Hurricane Ida flood (south of New Orleans). (c) Computed MNDWI and (d) categorized MNDWI derived from Sentinel-2 imagery after the Hurricane Harvey flood (east of Houston).

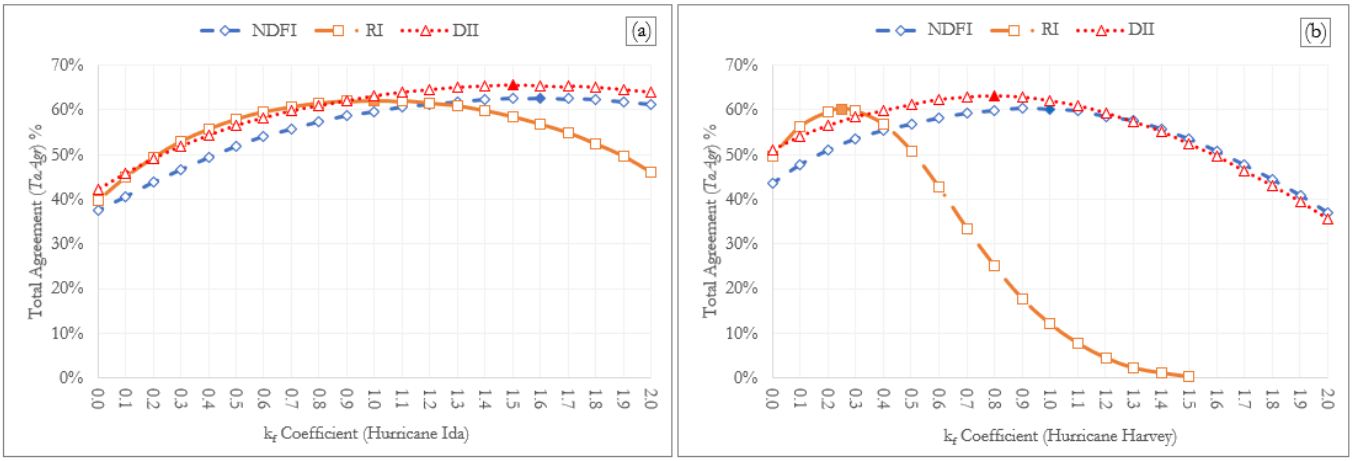


Fig. 7. Examining different k_f coefficients in (4) and assessing total agreement percentage using (7) to calibrate k_f coefficient and find optimal flood thresholding values for NDFI, RI, and DII images related to (a) Hurricane Ida flood, south of New Orleans and (b) Hurricane Harvey flood, east of Houston. The selected k_f threshold (peak agreement) is depicted with color-filled shapes (refer to Table II).

[see Fig. 7(a) and (8b)]. Finally, we applied a value of $k_f = 1.5$ (Thr = 0.6003) on the DII based on Fig. 7(a) and illustrate flooded areas in Fig. 8(c).

Furthermore, for extracting the flooded extent of the 2017 Hurricane Harvey to the east of Houston, Fig. 7(b) shows that $k_f = 1$ for the NDFI image has the best total agreement percentage, resulting in a threshold value equal to 0.4 used on the NDFI image to extract the flooded area [see Fig. 8(d)]. Next, we consider $k_f = 0.25$ [see Fig. 7(b)] and a threshold value equal to 0.002 was used to delineate flooded pixels from the RI image [see Fig. 8(e)]. Finally, $k_f = 0.8$ [see Fig. 7(b)], and a threshold value equal to 0.648 was used to map the flood from the DII image [see Fig. 8(f)].

We compared the flood extent of MNDWI and SAR change detection indices (NDFI, RI, and DII) in Fig. 9. These images certify visually that the flood extent maps using SAR indices and optical index are in good agreement for both Hurricane Ida and Hurricane Harvey, and flood regions are in the same locations and have a similar flood extent where the optical imagery is not covered by clouds. Moreover, Fig. 9 depicts that

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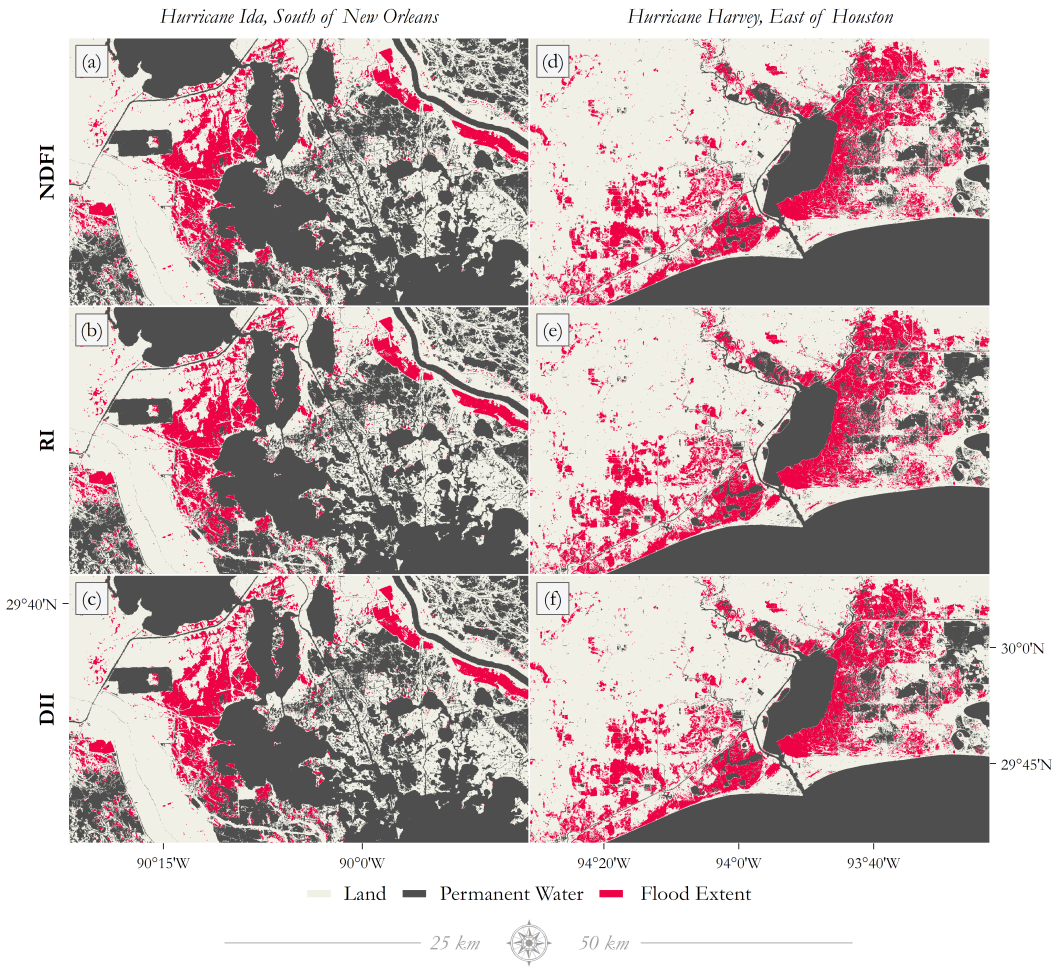


Fig. 8. Estimated flood extent by applying (a) threshold = 0.5013 on NDFI, (b) threshold = 0.0119 on RI, and (c) threshold = 0.6003 on DII for the 2021 Hurricane Ida (south of New Orleans). Estimated flood extent using (d) threshold value equal to 0.4 on NDFI, (e) threshold value equal to 0.002 on RI, and (f) threshold value equal to 0.648 on DII for the 2017 Hurricane Harvey (east of Houston).

the three SAR indices (DII/NDFI/RI) produce nearly identical flood extents after threshold calibration.

Fig. 10(a) summarizes the flooded area extracted by NDFI, RI, and DII, and highlights the agreement ratio with the MNDWI flooded area for the 2021 Hurricane Ida. We exclude under-cloud cover pixels from the SAR indices image to compare the results with MNDWI results (see Fig. 10(a) and Table II). Fig. 10(a) indicates that the NDFI flooded areas overlap with 77% of the MNDWI flood domain (46.90 out of 61.08 km^2). The agreement percentages for RI and DII with MNDWI flood are 77% and 80%, respectively [see Fig. 10(c)]. We also survey the agreement between SAR and optical image flood maps for Hurricane Harvey [see Fig. 10(b)]. Since the optical image has a cloud cover, we excluded it from the SAR images and then compared the estimated flood extents from optical and SAR images. Fig. 10(b) and Table II show that all three SAR indices estimate flood extent in accordance with vetted optical imagery flood maps due to the high agreement ratio. The agreement percentages are 78%, 80%, and 80% for NDFI, RI, and DII, respectively [see Fig. 10(d)]. These percentages suggest that the SAR change detection and thresholding method provides a good correlation and is well-matched with MNDWI results. Despite the lag between SAR and optical imagery for Hurricane Ida (seven days), agreement percentages were comparable to those of Hurricane Harvey, which had only a one-day lag. However, during other flood events with differing drainage systems, the lag time could introduce more uncertainty.

The optical image used here for calibration is affected by cloud interference, and (8) estimates the percentage of the flood area that SAR data can delineate under the cloud cover. Table II shows that more than 13% of Landsat 8 imagery after Hurricane Ida and 40% of Sentinel-2 imagery after Hurricane Harvey are covered by clouds (UndCldFld), which are captured using the SAR flood delineation technique.

IV. Discussion

Floods are one of the primary natural disasters that threaten human life and assets. The devastating effects of floods have prompted responsible authorities to pay critical attention to this hazard in recent decades. Providing flood extent maps during and after flood events will play a significant role in reducing the catastrophic consequences of floods.

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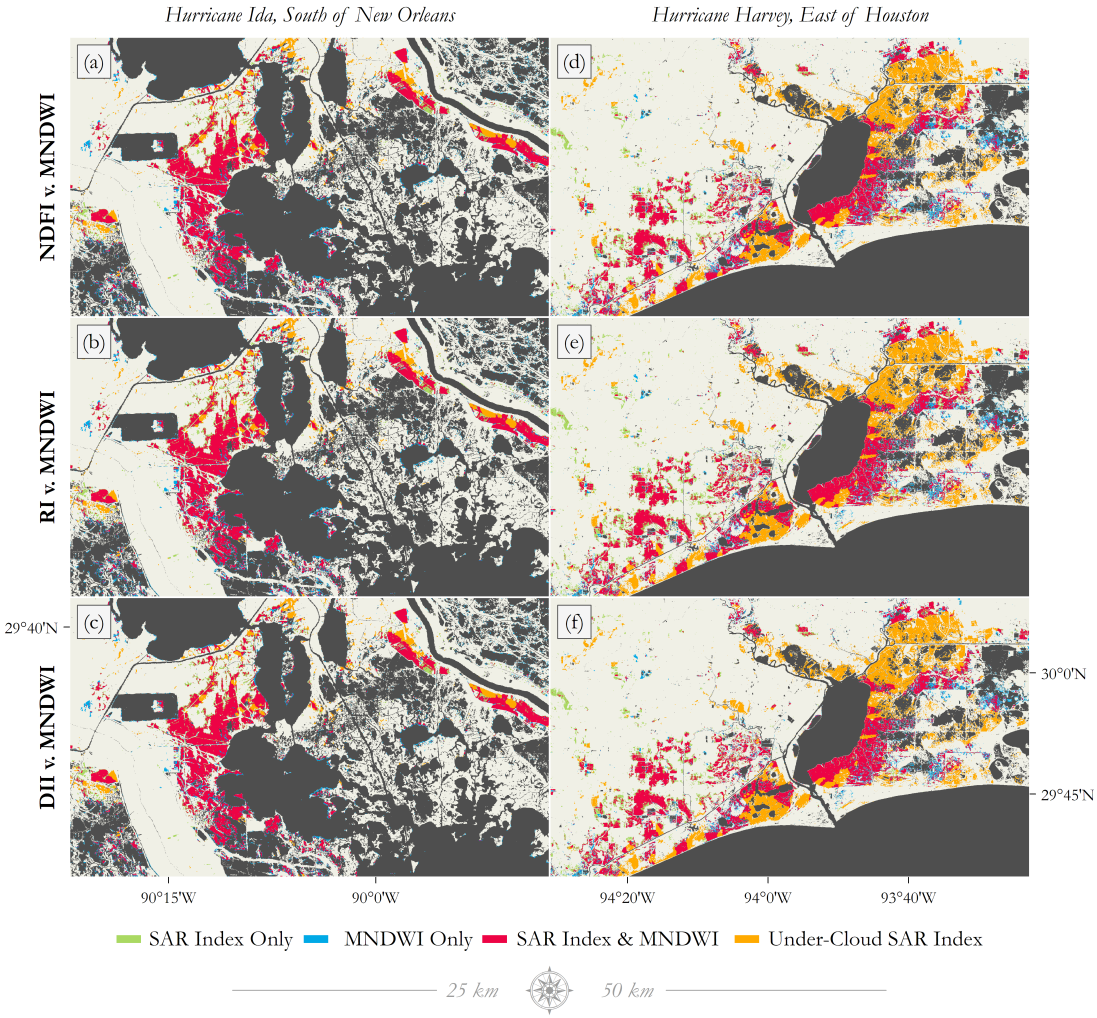


Fig. 9. Cross-comparison of MNDWI-based flood extent and (a) NDFI, (b) RI, and (c) DII flood extents for Hurricane Ida (south of New Orleans). Cross-comparison of MNDWI-based flood extent and (d) NDFI, (e) RI, and (f) DII flood extents for Hurricane Harvey (east of Houston).

TABLE II
LIST OF THRESHOLDS VALUES (Thr), k_f COEFFICIENTS, AGREEMENT (Agr) AND TOTAL AGREEMENT (ToAgr) PERCENTAGES, AND UNDER CLOUD FLOOD (UndCldFld)

	Hurricane Ida, South of New Orleans					Hurricane Harvey, East of Houston				
Indices	k_f	*Thr*	*Agr*	*ToAgr*	*UndCldFld*	k_f	*Thr*	*Agr*	*ToAgr*	*UndCldFld*
NDFI	1.60	0.5013	77%	63%	14%	1.00	0.4000	78%	60%	40%
RI	1.00	0.0119	77%	62%	14%	0.25	0.0020	80%	60%	40%
DII	1.50	0.6003	80%	66%	13%	0.80	0.6480	80%	63%	40%

Organizations such as the Global Flood Partnership (GFP; https://gfp.jrc.ec.europa.eu) and SERVIR (a joint development initiative between NASA and the U.S. Agency for International Development) are multidisciplinary organizations that focus on flood risk management and the development of efficient and effective flood monitoring tools. Models such as the one presented here provide critical information during flood events and can be deployed rapidly during calls for action by agencies such as GFP and SERVIR. Moreover, this study can easily integrate with other tools developed within GEE (e.g., FwDET-GEE, a floodwater depth estimation model [82]). FwDET-GEE requires only a flood extent layer as input, which this study provides. Flood extent and depth estimation can further be combined with datasets such as population, agriculture, and land-use/land-cover to assess the impacts of floods on humans and the environment, and prepare for postflood aid.

This study surveys flood extents using multisource and multitemporal data via change detection and thresholding

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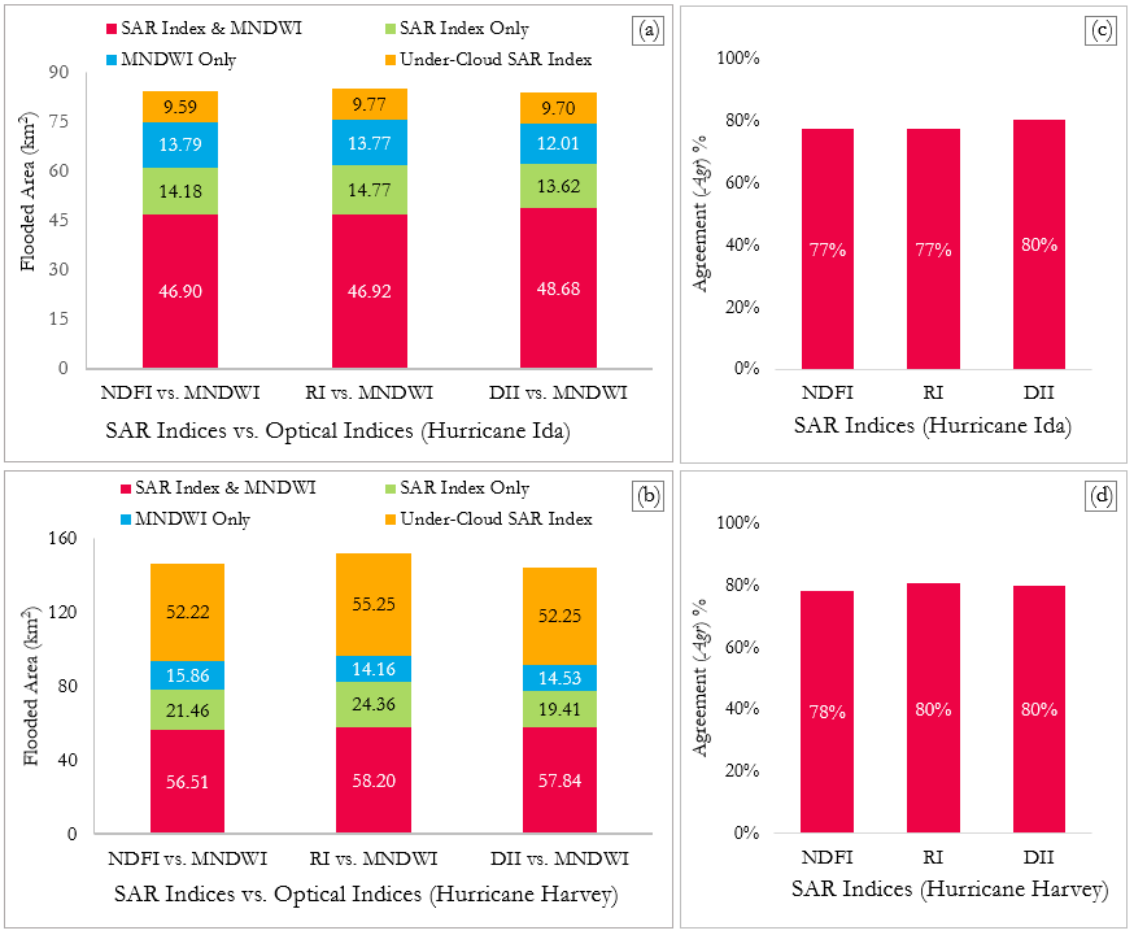


Fig. 10. Area in the agreement between MNDWI-based flood extent and NDFI, RI, and DII flood extents for (a) Hurricane Ida and (b) Hurricane Harvey. SAR indices agreement percentage: (c) Hurricane Ida and (d) Hurricane Harvey.

techniques on GEE. All steps for the flood delineation procedure (see Fig. 1) were completed in a couple of minutes using the geoprocessing tools and cloud-computing capabilities of GEE, without needing to download the data. We utilized the NDFI, RI, and DII SAR indices for flood delineation and validated results with available optical images. Previous studies considered a constant k_f coefficient equal to 1.5 in (4) to extract flooded pixels; however, our study showed that this value is sensitive to different indices and study areas. We examined different values for the k_f coefficient and assessed the total agreement percentage. We, therefore, considered the largest total agreement percentage for deriving suitable threshold values and extracting flooded pixels using SAR indices images. It seems that a universal thresholding technique for change detection flood mapping will require more research and robust statistical analysis; however, it may be the case that thresholding relative to the geographies and data used will often be the superior approach.

A major factor in the time it takes to produce flood maps is the preprocessing and postprocessing of images on local personal computers. To highlight the speed of this heuristic, the threshold determination script for Hurricane Harvey was completed in only 2 min and 17 s, and the flood extent script was completed after 1 min and 10 s; however, run times will vary based on the size of the study area, the number of images, spatial/temporal resolution of the data, and various constraints on the Earth Engine backend. Traditional methods of flood mapping are often time-consuming and require specialized software knowledge. Practitioners need to search SAR data repositories online, find the best images for the study area, download and store them on personal computers, and preprocess the data using software such as SNAP (The Sentinel Application Platform) [104]. At this stage, SAR images are ready for further analysis, such as calculating the SAR indices using local hardware and software (e.g., ArcGIS and QGIS). GEE expedites all these preprocessing and postprocessing steps. Furthermore, changing the study area requires only delineating a new polygon on the web coding interface.

We evaluated the pixel values of each SAR index, which showed that NDFI and DII are both well distributed across the two study areas; however, the RI pixel values were not well distributed, even after conducting a focal smoothing filter. An investigation of outlying RI values on the high end of the distribution showed that these anomalies may be primarily associated with water structures, such as traveling ships and cargo containers along the water edge. Relatedly, SAR histogram outliers have been used to detect ships in ice-infested waters [105], [106]. The distribution of RI

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values highlights the challenges associated with specifying a threshold value when using RI to detect flood inundation, a problem that was also noted by Vanama et al. [3]. More research into SAR-based index outliers is warranted; however, the bulk of the outliers identified was situated along permanent water bodies, which were ultimately masked out as part of the heuristic devised here. We removed any remaining outliers in the RI visualizations to more effectively smooth the rescaled version of the index ranging from 0 to 1. While this study focused on rural coastal areas, it is important to note that potential outliers from reflective structures may pose greater challenges in flooded areas that are more densely developed. In this respect, RI may be a useful index to identify outliers that emerge from SAR derivations. Conversely, an index that has normalization as part of its formulation, such as NDFI, may be useful to prevent outlier emergence though this could also be viewed as artificial masking of anomalous data.

Our results (see Fig. 5) demonstrate that NDFI and DII are reliable options for flood mapping both precalibration and postcalibration, whereas RI failed to extract floods for Hurricane Harvey without first being calibrated. This study validates the generalizability and simplicity of NDFI for emergency flood mapping, as has been attested in other studies [4], [40], [89], [90], and also indicates that DII can have similar or better flood extent results (refer to Table II). Combining SAR indices also warrants further research to evaluate if, for example, NDFI and RII work better than NDFI or RII individually.

Since these indices are calculated relative to available before/after imagery, which can have variable date range intervals, they are subject to the modifiable temporal unit problem (MTUP [100]). Any NDFI, RII, or DII calculation will vary depending on the image dates selected, and there may be errors incurred if unsuitable images are used in the calculation. Cross-comparisons can also be a challenge if there is a lack of consistency in the date range intervals. Despite these issues of time, Sentinel-1 is temporally continuous and at a finer temporal resolution (6/12-day revisit rate) than other SAR predecessors (e.g., PALSAR; 46-day revisit rate [101]), and a great deal of information can be extracted from any two before and after images. Moreover, since SAR data penetrate the cloud layer, there is less variation in the temporal ranges of before/after imagery than with optical platforms where the time problem presents a more considerable hurdle.

The limited availability of cloud-free optical images during and after floods can be overcome by leveraging SAR data for flood delineation. Although we utilize Landsat 8 and Sentinel-2 optical images for flood extent validation, these optical images are affected by cloud cover. For instance, 9.59 km^2 of flooded areas (14%) from Hurricane Ida that was mapped using NDFI are missed by MNDWI because of the abundance of cloud cover and the inability of optical imagery to penetrate the cloud layer. The condition is more critical for the Sentinel-2 image after the 2017 Hurricane Harvey flood, where 52.22 km^2 of flooded areas (40%) are missed in the optical image (NDFI versus MNDWI; see Figs. 9 and 10). This highlights the strength of SAR data, which can extract floods in all weather and during any time of day/night.

Although there is no perfectly accurate reference data available, collecting highly reliable reference information is of great significance to assess every research study's results [40], [107], [108]. The lack of ground-truth data is a well-known challenge in extracting floodwater from remote sensing data, and the validation procedure that we adopted here has its limitations. The results of both methods, which are summarized in Fig. 10, elucidate differences between the SAR and optical methods and highlight potential errors across each system. The disagreement area for NDFI is 27.97 km^2 (13.79 + 14.18), which is 60% of the common area in both SAR and optical (27/46.9 $\times$ 100). There are many potential sources of error associated with optical images, such as those introduced by atmospheric conditions, solar illumination, soil moisture, and vegetation phenological changes [102], [109], [110]. In addition, optical images are susceptible to errors such as classifying cloud shadows as water and the inability to map the flood under vegetation, canopies, and debris accumulations [102]. Furthermore, for floodwater delineation based on MNDWI, we set a threshold value to differentiate floodwater verses nonflood regions; this technique may lead to underestimation or overestimation of floodwater extents, as may be the case with any thresholding technique. As for SAR, there are issues associated with tarmac surfaces (e.g., airports and roads) being misclassified as water, floodwater being missed during wind and heavy rain (roughening the water surface), and floods occurring under vegetation and in heavily built-up areas (due to problems associated with double-bounce backscattering signatures) [40]. These shortcomings inherent to the usage of SAR data in flood mapping are addressed in part by our multisource remote sensing approach. By using multiple lines of evidence (i.e., optical imagery and multiple SAR indices), we can overcome some of the problems associated with errors of omission. While combining datasets can introduce errors of commission, it is arguable that, in many applications, overestimating flood extent will be preferable as you would be less likely to miss affected people and places. Moreover, the error percentage (or nonmatching optical and SAR flooded area) has been shown to increase when using different image resolutions, variable image acquisition times, and differing flood extent detection techniques [40], [43], [111]. For example, in this study, although Sentinel-1 images are available within a few hours of the satellite pass, it has a temporal resolution of six or 12 days depending on the location of the study area. This temporal resolution could miss unpredictable dynamics of flood events and fail to observe floods during critical conditions. Therefore, other techniques and data are essential to have daily or even higher temporal resolution information for flood emergency management. Utilizing multiple images and ground-truth data (e.g., high watermark collections) can help reduce the uncertainties in validating flood extents.

VH polarization has been widely used for reliable flood mapping [112], [113], [114], [115], and consistent with these studies, we used Sentinel-1 C-band VH polarization data for flood water estimation. However, all HH, VV, HV, and VH polarizations of SAR data, as well as coupled polarization,

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have been used in the literature for flood mapping [112], [113], [114], [115]. Manjusree et al. [112] showed HH polarization works well for surface water detection. In addition, their study demonstrated that VV polarization has a slight advantage compared to VH polarization for flood delineation; however, surface water roughness generated by rains or winds can affect VV polarization significantly. As an area of future research, it is worth mentioning that, in the GEE code that we provide [98], users can easily switch between different polarization bands and quickly test the results.

V. Conclusion

This study provides a new multisource and multitemporal image application for fast flood extent mapping in coastal areas leveraging the capabilities of GEE. This article brings together optical flood detection, SAR-based flood detection techniques, geographic philosophy (i.e., considering modifiable areal unit problems and variability in localized analyses), and automated thresholding techniques in a way that has not been done before on a platform that is growing in popularity. The process is fast, which will assist event response, and researchers only need to engage minimally with the code to produce reliable and geographically generalizable outputs. We have also included scripts in a Harvard Dataverse publication [98] that facilitate threshold automation to enhance the replicability and reproducibility of the work.

We surveyed the coastal floods of the 2017 Hurricane Harvey and the 2021 Hurricane Ida, which are two of the strongest natural disasters in recent years. To the best of our knowledge, this is the first application of SAR flood detection for Hurricane Ida. The flood extent maps obtained from this method demonstrated a very good agreement with flood maps derived from available Landsat and Sentinel-2 optical images, and all the agreement percentages are above 77%. These accurate and near real-time flood maps can assist policymakers and government agencies in mitigating natural hazards and make it possible for planners and disaster-response agencies to evaluate the hazard risk and respond efficiently and immediately to the potential threat of flooding.

This study applied a technique using multitemporal SAR change detection and thresholding indices, along with optical imagery indices, for flood delineation and then analyzed the resulting pixel classifications. We showed that a constant thresholding value could fail in some cases and then adapt multisource and multitemporal techniques to automate the process of finding flooded pixels using SAR and optical indices. Moreover, this study highlights the significant capabilities of GEE to provide automated, rapid, and efficient flood maps, as well as multidisciplinary information to assess further supplementary intersect datasets, such as exposed populations and affected agricultural and urban areas. Further research in this context is critical for a wide range of responsible authorities. Also, since it is hard to find optical imagery without cloud cover for the same date of SAR data, simulating flood extent using a calibrated hydrodynamic model would be another alternative for validating the SAR-based flood extent map.

Data Availability

Sentinel-1 and Sentinel-2 are public domain datasets provided by the ESA and can be accessed from the Copernicus Data and Information Access Service (DIAS): https://sentinel.esa.int/web/sentinel/sentinel-data-access. Landsat imagery is public domain data supplied by the USGS as part of the National Land Imaging (NLI) program: https://www.usgs.gov/Landsat-missions. Sentinel-1, Sentinel-2, Landsat, and the JRC water body data are also available through the Google Earth Engine data catalog: https://developers.google.com/earth-engine/datasets.

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